

NBA Shooting Streaks: Hot Hand or Chance?

- Question: Is making 5 shots in a row evidence of a “hot hand”, or could it be attributed to chance?
- Null Model: We assume the shots are independent with a constant make probability, p .
- Class Connection: This is a Monte-Carlo simulation using RNG. This generates random outcomes many times to estimate probabilities.



Setup

- n = 100 shots per simulation
- 20,000 simulations per p
- p = 0.35 p = 0.45

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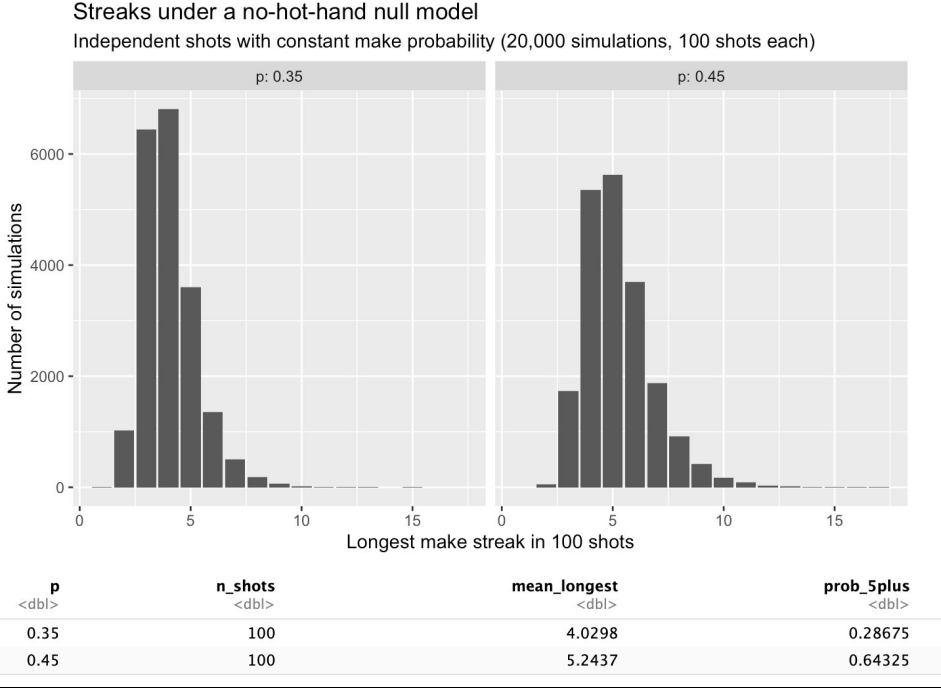
- Mean longest streak ≈ 4.03
- P(streak ≥ 5) ≈ 0.287 (≈29%)

p = 0.45

- Mean longest streak ≈ 5.24
- P(streak ≥ 5) ≈ 0.643 (≈64%)

What does this mean?

Under this no-hot-hand model, meaning every shot is independent, there is a 29% chance that a streak of 5+ threes, and a 64% chance that a streak of 5+ twos can be attributed to luck. So a streak of 5+ makes is not evidence of a hot hand, it is just what randomness looks like.



Takeaways

- Streaks happen naturally under randomness.
- A streak of 5+ makes is not rare ($\approx 29\%$ for 3pt, $\approx 64\%$ for 2pt).
- This model shows that streaks alone are weak evidence for a hot hand.

Possible Extensions

- True hot hand model
 - After each make p would increase temporarily, making the chance of a streak of 5+ makes occurring increase every time a shot goes in.
 - If the hot hand truly existed, then we would see longer streaks and more total 5+ streaks than we did in the null model.